

NIRAJ SINGH

Software Quality Engineering

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PROFESSIONAL SUMMARY

Senior SDET with 10+ years of experience defining quality assurance strategy and leading delivery across Financial Services, Retail, and E-Commerce. Specialises in building scalable test automation across Web, API, microservices, and data layers within cloud-native environments, with a growing focus on AI quality engineering including agentic system evaluation, LLM output validation, and MCP-integrated tooling. Proven record of QA governance at programme level, risk-based testing, cross-team mentoring, and quality embedding through shift-left practices and CI/CD-integrated pipelines.

TECHNICAL SKILLS

Test Automation & Frameworks: Playwright, WebDriverIO, Selenium, Appium, REST Assured, Karate, JUnit, Pact (Contract Testing), WireMock, Cucumber/Gherkin, Pytest, Percy

Programming Languages: Java, TypeScript/JavaScript, Python

Cloud & DevOps: GCP, Docker, Kubernetes, Azure DevOps, Jenkins, GitHub Actions

Data: SQL, BigQuery, Spanner, MongoDB, Kafka, Pub/Sub

Performance & Security: Gatling, OWASP ZAP

Observability: Dynatrace, Splunk

Accessibility: WCAG 2.1 Guidelines, Axe-core, NVDA

AI & Quality Engineering: DeepEval, RAGAS, LLM-as-judge, GitHub Copilot, Claude AI, MCP (Atlassian, Playwright), OpenAI Agents SDK, LangGraph, HuggingFace, Prompt Engineering

Tools & Methodologies: Jira, X-ray, BrowserStack, Postman, Git, Gradle, Maven, Agile/Scrum, Trunk-based Development, BDD/TDD

PERSONAL PROJECTS

AI Quality Engineering Lab

- Built and deployed an alter-ego AI agent on HuggingFace, a conversational agent trained on my professional expertise to answer questions about my experience, skills, and career, doubling as a live testbed for agentic architecture and LLM evaluation techniques.
- Designed an LLM evaluation harness using DeepEval and RAGAS to measure hallucination rates, answer relevance, faithfulness, and groundedness against defined acceptance thresholds; applied LLM-as-judge scoring to assess the quality and consistency of chatbot responses.
- Integrated a GitHub Actions CI pipeline running RAGAS eval thresholds as automated pass/fail quality gates.

PROFESSIONAL EXPERIENCE

Publicis Sapient | Oct 2014 – Present

Client: Sonepar (B2B E-Commerce), London

Senior Quality Engineer Aug 2024 – Present

- Owned QA across 2 feature teams, setting test strategy, running sprint ceremonies, and holding release sign-off authority.
- Designed and integrated agentic AI workflows into the QE practice using MCP (Atlassian and Playwright), GitHub Copilot, and custom-built agents and skills, automating test plan generation, X-Ray test case creation from Jira requirements, and self-healing tests, cutting test creation and maintenance effort by 30%.
- Built a Playwright TypeScript framework from scratch using Page Object Model architecture, replacing an unstable legacy suite.
- Defined a test grading strategy (Golden Path and Critical suites), reducing execution time by 70%+ while maintaining near 100% pass rates.
- Enhanced the WebDriverIO regression suite, reducing execution time by 20% and achieving a 90%+ pass rate.
- Introduced session-based exploratory testing as the standard validation approach for new features, moving the team beyond scripted automation coverage.
- Shifted coverage toward unit and component layers, reducing E2E suite size while improving stability and execution speed.

Client: Lloyds Banking Group, London**Senior Quality Engineer** Feb 2020 – Jul 2024

- Led QA strategy and delivery for a GCP-based Customer Data Platform (CSoR and SOI), handling millions of customer records across data migration and data warehouse workstreams.
- Owned QA governance for the programme, producing test plans, risk logs, and progress reports for steering group review.
- Built a Java/REST-Assured framework with Pact for contract testing, automating API and Kafka event streaming tests across microservices deployed with WireMock service virtualisation.
- Built a Gatling-based performance testing framework for microservices load testing.
- Used Dynatrace for distributed tracing and post-release production monitoring to pinpoint bottlenecks and validate SLAs in real time.
- Embedded OWASP ZAP into the CI pipeline for shift-left security testing, catching vulnerabilities at build time across all microservices.
- Designed automated tests for dbt jobs validating data integrity and transformation accuracy in BigQuery, replacing entirely manual validation across large-scale datasets.
- Mentored engineers across feature teams on BDD and framework adoption.

Client: Academy Sports & Outdoors, India**Senior Quality Engineer** May 2019 – Jan 2020

- Led a 3-member QA team, delivering in-sprint automation using Selenium/Java for new features across cart and checkout, lifting regression pass rate from 60% to 80%.

Client: Bed Bath & Beyond, India**Quality Engineer** Sep 2016 – Apr 2019

- Built an end-to-end UI regression suite for web and mobile search using Selenium and Appium/Java.
- Owned story grooming and acceptance criteria definition for the QA workstream.

Client: Research Navigator (Cambridge Associates), India**Quality Engineer** Oct 2014 – Sep 2016

- Sole QA engineer; built QA processes from the ground up and owned end-to-end manual and exploratory testing for a financial research platform.

EDUCATION & CERTIFICATIONS

Bachelor of Engineering, Army Institute of Technology, Pune | 2010 – 2014

Introduction to Generative AI, Google Cloud | 2024

Google Prompting Essentials | 2026